3.6 Load Current Detection for CT Installation and Wiring

3.6.1 CT Selection

1. The appropriate rated ratio of primary to secondary current shall be determined. The primary current is recommended to be 1.5*In (the actual rated current).
2. The rated voltage is more than or equal to the system voltage.
3. The selection for secondary current should be 5A or 1A, 5A is suggested.
4. CT accuracy shall be class 0.5 or 1.0, class 0.5 is suggested
5. The nominal secondary capacity (rated load) of the CT shall meet the requirement of secondary impedance (≥ 10VA when the secondary current is 5A). The capacity and the maximum one-way wiring length from the CT to the active power filter shall be calculated according to the following formula:

$$L_{max} = \frac{P_{CT} - P1}{I^2} \cdot \frac{S}{\rho} \cdot \frac{1}{2}$$

Wherein:

L_{max}	is the maximum one-way wiring length from the CT to the system cabinet (m);
P_{CT}	is the nominal secondary capacity of the CT (VA);
P1	is the capacity loss and the internal impedance of the system cabinet (each module's internal loss is around 2 VA);
I	is the secondary current of the CT (A);
S	is the cross-section area of the copper conductor (mm^2);
ρ	is the resistivity of the copper conductor (calculated according to 0.0178 Ω × m/ mm^2);